

09_Design_System.md

Part 2 — Components & Interaction System

Component Philosophy

Every component must be:

- Reusable
- Consistent
- Accessible
- Responsive
- Token-driven
- Independent

Components should solve one problem only.

Component Hierarchy

Design Tokens

↓

Atoms

↓

Molecules

↓

Organisms

↓

Templates

↓

Pages

Atoms

Smallest reusable elements.

Components:

- Button
- Icon
- Text

- Badge
- Avatar
- Input
- Label
- Divider
- Spinner
- Tooltip

Rules:

- No business logic
 - Highly reusable
 - Token-based styling
-

Molecules

Combination of multiple atoms.

Components:

- Search Bar
- Stock Card
- Statistic Card
- Navigation Item
- Notification Item
- Form Field
- AI Suggestion Card

Rules:

- Single responsibility
 - Independent
 - Reusable across modules
-

Organisms

Complete functional sections.

Components:

- Navigation Bar
- Sidebar
- Hero Section
- Portfolio Section
- Watchlist Panel
- Market Table
- Leaderboard

- Course List
- AI Chat Panel

Rules:

- Can contain multiple molecules
 - Independent section
 - Responsive
-

Templates

Page layouts without business data.

Templates:

- Dashboard Layout
 - Market Layout
 - Authentication Layout
 - Portfolio Layout
 - Learning Layout
 - Admin Layout
-

Page Structure

Each page contains:

- Header
 - Navigation
 - Main Content
 - Secondary Panel (Optional)
 - Footer (Public Pages)
-

Button System

Variants

- Primary
- Secondary
- Outline
- Ghost
- Text
- Danger
- Success

States

- Default
- Hover
- Active
- Focus
- Disabled
- Loading

Input System

Supported Inputs

- Text
- Email
- Password
- Search
- Number
- Date
- Dropdown
- Checkbox
- Radio
- Toggle
- Textarea

States

- Default
- Focus
- Filled
- Error
- Disabled
- Success

Card System

Card Types

- Stock Card
- Portfolio Card
- Analytics Card
- AI Card
- Course Card
- Competition Card
- Achievement Card

Rules

- Consistent padding
 - Rounded corners
 - Soft elevation
 - Hover feedback
-

Navigation Components

Primary Navigation

- Sidebar (Desktop)
- Bottom Navigation (Mobile)

Secondary Navigation

- Tabs
 - Breadcrumbs
 - Filters
-

Modal System

Modal Types

- Confirmation
- Success
- Error
- Information
- AI Response
- Trade Confirmation

Rules

- Background blur
 - ESC to close
 - Click outside (optional)
 - Focus trapped inside
-

Table System

Used For

- Market Data
- Portfolio
- Trade History

- Leaderboards

Features

- Sorting
 - Filtering
 - Pagination
 - Responsive Layout
-

Chart System

Chart Types

- Line Chart
- Candlestick Chart
- Area Chart
- Bar Chart
- Pie Chart

Rules

- Interactive
 - Responsive
 - Accessible
 - Smooth animations
-

Feedback Components

Success

Warning

Error

Info

Loading

Empty State

Skeleton Loader

Toast Notification

Interaction Principles

- Every clickable element provides visual feedback.
 - Hover effects should be subtle.
 - Focus states are mandatory.
 - Animations should communicate state changes.
 - Avoid unnecessary motion.
-

Responsive Behaviour

Desktop

- Sidebar
- Multi-column layout

Tablet

- Collapsible navigation
- Adaptive grid

Mobile

- Bottom navigation
 - Single-column layout
 - Floating Action Button
-

Accessibility Rules

- Keyboard navigation supported
 - Focus indicators visible
 - Minimum touch target: 44×44 px
 - Semantic HTML
 - ARIA labels where required
-

Component Naming Convention

Format

ComponentName

Examples

Button

StockCard

PortfolioSummary

AIChatPanel

MarketTable

LeaderboardCard

Component Validation Checklist

- Reusable
- Responsive
- Accessible
- Token-based
- Documented
- Tested
- Performance optimized

Related Documents

- 08_Information_Architecture.md
- 10_API_Architecture.md
- 12_Engineering_Standards.md

Approval Status

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Next Document

09_Design_System.md — Part 3